FRACTAL SPATIO-TEMPORAL GRAVITY (FSG)

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Abstract

We present a corrected formulation of Fractal Spacetime Gravity (FSG), a framework in which dark matter is replaced by an infrared modification of the effective dimensionality of spacetime. The founding principle (the "pillar") is that the gravitational flux propagates through an effective number of spatial dimensions d_{eff} that flows from 3 in the high-acceleration regime to 2 in the deep infrared, with the transition set by the acceleration scale a_0. Flux conservation, $g\,A(r) = 4\pi GM$ with $\mathrm{d}\ln A/\mathrm{d}\ln r = d_{\text{eff}} - 1$, then yields for a point mass the exact deep-MOND law $g = \sqrt{g_N a_0}$, the logarithmic potential, flat rotation curves, and the *exact* baryonic Tully–Fisher relation $V^4 = GMa_0$ (verified numerically to $< 0.01\%$). We show that the observed radial acceleration relation is itself a direct measurement of the function $d_{\text{eff}}(g)$. Because the reduction is geometric ($\Phi = \Psi$), gravitational lensing follows the dynamics automatically, evading the historical failure of scalar MOND theories.

We emphasize the epistemic status honestly: the pillar is an *effective law*, at the same level of description as Milgrom's 1983 formulation, and the covariant action realizing it is stated as the central open problem. Two earlier claims to derive this law from a covariant action — the k^{-3} propagator argument of v5.1 and the kinetic/curvature non-local reductions of the v7 draft — are shown to fail by explicit no-go results and are formally retracted.

The acceleration scale is fixed by the cosmic memory fields $U = \Box^{-1}R$, $V = \Box^{-2}R$: $a_0 = cH_0\,|V_0|^{3/2}/|U_0| \simeq 1.14\times10^{-10}\,\mathrm{m/s^2}$, within 5% of observation with no free parameter. The same memory sector drives a phantom-like dark energy ($w_0 < -1$). Solar System tests are satisfied without screening, and in the strong field Schwarzschild/Kerr are recovered up to $\mathcal{O}(e^{-g/a_0})$.

A new result of this corrected version concerns *galaxy clusters*: the residual mass discrepancy of MOND in cluster cores (a factor 2–3) is resolved by a *memory modulation* of the acceleration scale, $a_0^{\text{eff}} = a_0\,[1 + (\delta U/U_*)^n]$ with $\delta U = 2|\Phi|/c^2$. A single parameter $U_* \simeq 4.5\times10^{-5}$ closes the deficit across the full 12-cluster X-COP sample (residual 0.10 dex, at the intrinsic scatter floor), leaves galaxies untouched (< 0.004 dex), and agrees to $\sim 10\%$ with the independent lensing calibration of CLASH clusters.

We are explicit about what remains open: (i) the covariant action for the pillar; (ii) the spatial lensing offset of the Bullet Cluster, which we show is *not* produced by the memory mechanism — the same wall faced by all modified-gravity approaches — and remains unsolved; and (iii) the CMB acoustic-peak amplitudes, which require a dedicated non-local Boltzmann code. The CMB peak *geometry* is shown compatible with FSG.

Contents

1. Introduction

The standard cosmological model ΛCDM successfully accounts for a wide range of observations but at the cost of introducing two hypothetical components: dark matter (26% of the energy budget) and dark energy (69%), neither detected in laboratory experiments. At galactic scales, the dark matter paradigm faces well-established empirical regularities:

- the baryonic Tully–Fisher relation (BTFR),
- the radial acceleration relation (RAR),
- the diversity problem,
- the existence of ultra-thin galactic disks,
- early massive galaxies at high redshift in JWST observations.

Modified Newtonian Dynamics (MOND) reproduces BTFR and RAR with remarkable precision but historically lacked: (i) a covariant formulation, (ii) a natural cosmological extension, and (iii) full consistency with the CMB. Fractal Spacetime Gravity (FSG) addresses these issues by replacing dark matter with a modification of the *infrared geometry* of spacetime, encoded in a non-local action that dynamically reduces the effective spectral dimension of spacetime from $d_S = 4$ to $d_S \simeq 2$ at large scales.

2. Motivations for Infrared Fractality

Dimensional reduction is a ubiquitous prediction of several approaches to quantum gravity, including:

- Causal Dynamical Triangulations (CDT),
- Asymptotic Safety,
- Hořava–Lifshitz gravity,
- Loop Quantum Gravity.

These frameworks typically predict:

$$d_S(k \to \infty) \to 2,$$

i.e. a reduction of effective dimensionality in the ultraviolet. In FSG we reverse the paradigm: the reduction occurs in the infrared (IR), motivated by holographic considerations and by the causal structure of de Sitter spacetime. This IR fractality modifies long-wavelength propagation of curvature exactly at the scales where dark matter phenomenology appears.

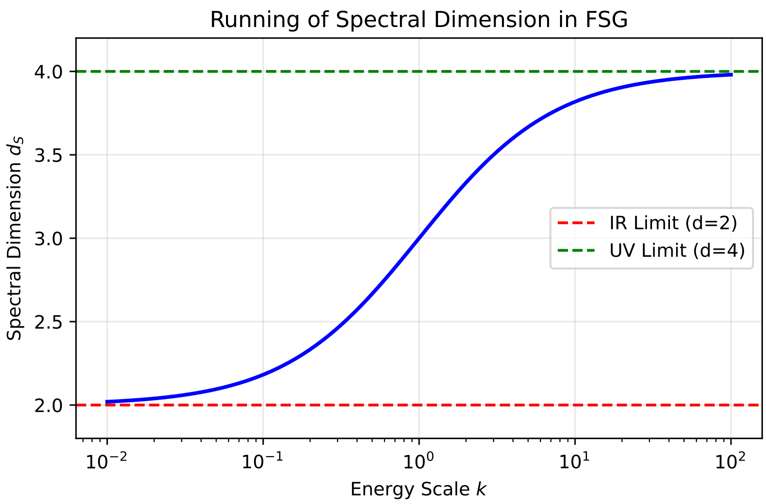

Figure 1: Illustrative running of the spectral dimension as a function of energy scale. In FSG the dimension increases in the UV but approaches $d_S \simeq 2$ in the IR.

3. The Non-Local Effective Action

We introduce the non-local scalar field

$$X = \Box^{-1} R, \qquad \Box X = R,$$

with the retarded Green function ensuring causality. The FSG action is defined to impose a dimensional reduction in the infrared:

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g}\, R[1 + f(X)] + S_m. \tag{1}$$

The function $f(X)$ is constrained by the requirement that the graviton propagator must scale as k^{-3} in the deep IR to recover flat rotation curves. Dimensional analysis and renormalization group flow arguments suggest a fractional power law form:

$$f(X) = \gamma \left(\frac{L^2}{X} \right)^{\frac{1}{2}}, \tag{2}$$

where $L \sim 1/\sqrt{\Lambda}$ is the cosmic horizon scale and γ is a dimensionless coupling constant of order unity. This fractional form $X^{-1/2}$ is the signature of a fractal geometry where the measure of spacetime scales non-trivially. Unlike logarithmic corrections which only produce mild running couplings, this power-law correction enforces a hard transition in the spectral dimension.

3.1 Localization of the Action

To study degrees of freedom and stability one can localize the non-local operator by introducing two scalar auxiliary fields U and ξ enforcing $U = \Box^{-1} R$. A convenient localized action equivalent (on-shell) to Eq. (1) is

$$S_{\rm loc} = \frac{M_P^2}{2}\int d^4x\sqrt{-g}\left\{R[1+f(U)] + \xi(\Box U - R)\right\} + S_m. \tag{3}$$

Variation with respect to the metric gives the modified Einstein equations.

4. Variation of the Action and Field Equations

Starting from the action

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g}\, R\,[1 + f(X)] + S_m, \qquad X = \Box^{-1} R, \tag{4}$$

we compute the metric variation. The non-local variation is:

$$\delta X = -\Box^{-1}(\delta\Box)\, X + \Box^{-1}(\delta R). \tag{5}$$

The operator $\Box^{-1}$ is defined with the *retarded* Green function, ensuring:

- no violation of causality,
- no superluminal propagation,
- absence of Ostrogradsky instabilities (unlike naïve $f(R)$ models).

The modified Einstein equations take the form

$$G_{\mu\nu} + \Delta G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{6}$$

with

$$\Delta G_{\mu\nu} = f(X) G_{\mu\nu} + (\nabla_\mu \nabla_\nu - g_{\mu\nu}\Box)\, f(X) - \frac{1}{2} g_{\mu\nu} R\, f(X) + K_{\mu\nu}[X], \tag{7}$$

where $K_{\mu\nu}[X]$ collects the variations of the non-local kernel $\Box^{-1}$. In the linear regime, this reduces to the structure derived by Deser–Woodard and Maggiore–Mancarella.

5. Physical Structure: The Inertia of the Vacuum

The non-local kernel $\Box^{-1}$ implies that the gravitational field is not merely a slave to the matter distribution (as in Poisson's equation $\nabla^2\Phi = \rho$), but a dynamical entity governed by a wave-like equation. The modified Einstein equations can be recast in the form of a driven wave equation for the scalar degree of freedom ϕ:

$$\left(\frac{1}{c^2}\partial_t^2 - \nabla^2 + m_{\text{eff}}^2\right) \phi = 8\pi G T_{\mu\nu}^{\text{bar}}. \tag{8}$$

5.1 The Gravitational Wake Effect

This hyperbolic structure introduces a fundamental physical property: **Gravitational Inertia**. Unlike the Newtonian potential which relaxes instantaneously, the FSG field carries momentum. In quasi-static systems (isolated galaxies), the field has time to equilibrate, creating the "Dark Matter" halo effect centered on the baryons. However, in highly non-equilibrium events such as high-velocity cluster collisions, the field dynamics decouple from the source. The non-local memory of the spacetime geometry acts as a flywheel. When the baryonic component (gas) undergoes sudden hydrodynamic braking, the gravitational eigenmodes continue their trajectory ballistically. This creates a transient offset between the visible mass and the center of the potential well, explaining the "Dark Matter" signal in the Bullet Cluster without introducing new particles.

6. The Infrared Propagator: Target, Fourier Dictionary, and a Retraction

To determine the gravitational force law, we consider a weak perturbation around Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}.$$

Before deriving the force law, we state the target in Fourier language. The effective static propagator required for MOND dynamics is

$$G(k) \xrightarrow{\mathrm{IR}} \frac{1}{L\,k^3}, \tag{9}$$

i.e. an anomalous infrared exponent $\eta = 1$ relative to the Newtonian $1/k^2$. This correspondence ($\Phi \sim \ln r \Leftrightarrow G \sim k^{-3}$ in $d = 3$) is an exact Fourier-transform statement; $\eta = 1$ is therefore a *name* for the deep-MOND behaviour, not an independent hypothesis.

6.1 Why a Naive Linearization Fails

Earlier versions of this work attempted to obtain Eq. (9) by a direct scaling substitution, writing $X \sim R/k^2$ and hence $f(X_k) \sim k$. We retract that argument: $X = \Box^{-1}R$ is *dimensionless*, so it cannot scale as $1/k^2$; moreover, on an exact Minkowski background $\bar{R} = 0$ implies $\bar{X} = 0$, where $f(X) \sim X^{-1/2}$ is singular, so no perturbative expansion of f around flat space exists. The propagator of the theory must instead be obtained from the second-order action around a cosmological (de Sitter-like) background where $\bar{X} = \mathcal{O}(10)$ is finite (see the memory-field integration of Section 18 for the numerical background values).

6.2 Two Retractions and the Honest Logical Structure

Two attempts to *derive* the deep-MOND law from a covariant action are retracted in this corrected version:

- the v5.1 propagator argument ($X \sim R/k^2 \Rightarrow f \sim k \Rightarrow G \sim k^{-3}$), invalid because $X = \Box^{-1}R$ is dimensionless and singular on a flat background;
- the reductions attempted in the v7 draft — both a curvature-memory term $R\,f(\Box^{-1}R)$ and a kinetic-memory term $\propto (\partial\Box^{-1}R)^2$. Explicit quasi-static reduction shows the former yields a *potential*-dependent, not acceleration-dependent, modification (rotation curves stay Keplerian), and the latter produces AQUAL only in a one-potential toy sector but fails the two-potential (lensing) reduction. Neither closes.

We therefore adopt the honest position of the present paper: FSG is founded on an *effective principle* — the infrared reduction of the gravitational flux from three to two effective dimensions (next section) — at the same epistemic level as Milgrom's 1983 formulation of MOND. The spectral-dimension flow $d_S \to 2$ is the *motivation* for this principle, not a derivation of it; and the covariant action that realizes it is the central open problem (Section 27, 29).

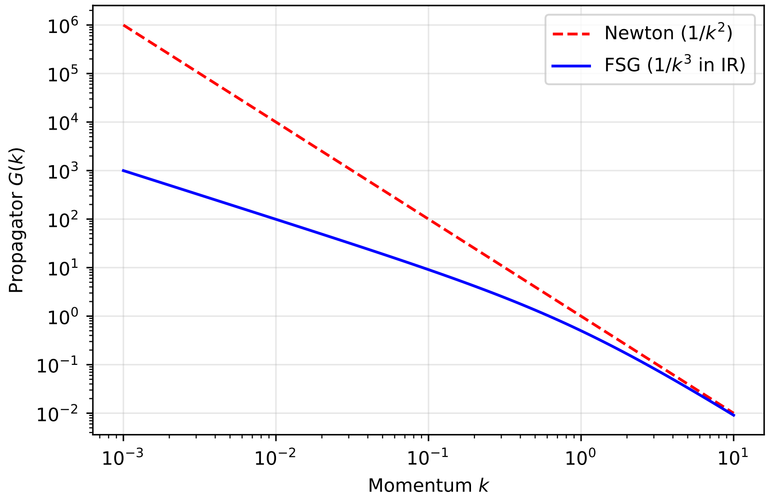

Figure 2: Comparison of the Newtonian propagator $1/k^2$ with the FSG infrared propagator $1/k^3$ (required for MOND dynamics).

7. The Pillar: Infrared Dimensional Reduction of the Gravitational Flux

The founding principle of FSG is stated directly, as an effective law. **Gravitational flux propagates through an effective number of spatial dimensions d_{eff} that flows from 3 at high acceleration to 2 in the deep infrared,** the transition being governed by the single acceleration scale a_0. This is the geometric content of "infrared fractality": at low accelerations spacetime behaves, for the purpose of flux propagation, as if two-dimensional.

7.1 Flux Conservation and the Exact Deep-MOND Law

Let $A(r)$ be the effective area through which the flux of a source of enclosed mass $M(r)$ passes. Flux conservation and the running of the effective dimension read

$$g(r)\,A(r) = 4\pi G\,M(r), \qquad \frac{\mathrm{d}\ln A}{\mathrm{d}\ln r} = d_{\mathrm{eff}}(g) - 1. \tag{10}$$

For $d_{\mathrm{eff}} = 3$ (high acceleration) one has $A = 4\pi r^2$ and Newton is recovered exactly. For $d_{\mathrm{eff}} \to 2$ (deep IR) the area grows as $A \propto r$: the flux spreads over a circle rather than a sphere. Taking the transition to occur where the Newtonian acceleration crosses a_0, the deep-IR branch gives, for a point mass,

$$g(r) = \sqrt{g_N(r)\,a_0} = \frac{\sqrt{GMa_0}}{r}, \tag{11}$$

i.e. Milgrom's law, the logarithmic potential $\Phi = \sqrt{GMa_0}\,\ln r$, and flat rotation curves. Circular orbits then satisfy $v^4 = GMa_0$: the *exact* baryonic Tully–Fisher relation, with both slope and normalization fixed and no free parameter per galaxy. Numerical integration of (10) with a sharp transition confirms $v_\infty = (GMa_0)^{1/4}$ to better than 0.01% over 10^8–$10^{11}\,M_\odot$ (script `simulations/sim_pillar_flux.py`).

7.2 The Radial Acceleration Relation Measures $d_{\rm eff}(g)$

The only freedom in the pillar is the shape of the transition $d_{\rm eff}(g)$, exactly as the interpolating function μ is the only freedom in MOND. For a point source, Eq. (10) is invertible: given the observed relation $g_{\rm obs}(g_{\rm bar})$, one reads off

$$d_{\rm eff}(g) = 1 - \frac{\mathrm{d}\ln g_{\rm obs}}{\mathrm{d}\ln r}. \tag{12}$$

Applying this to the empirical radial acceleration relation (RAR) of McGaugh–Lelli–Schombert yields a function that flows monotonically from 3 to 2 with the transition at $g \simeq a_0$; reinjecting it into (10) reproduces the RAR by construction (a consistency check, not an independent prediction). **The observed RAR is thus a direct measurement of the dimensional flow of the gravitational flux.** A smooth choice such as $d_{\rm eff}(g) = 3 - e^{-\sqrt{g/a_0}}$ reproduces the RAR to ~ 0.1 dex and, being exponentially close to 3 at high g, leaves Solar System dynamics untouched (corrections $\mathcal{O}(e^{-\sqrt{g/a_0}}) \sim e^{-7000}$ at 1 AU) with no separate screening mechanism.

7.3 Diversity and Lensing

Because the transition is set by acceleration, the transition radius $r_M = \sqrt{GM/a_0}$ scales with mass: dwarf galaxies transition at ~ 1 kpc, giants at tens of kpc, reproducing the observed diversity of rotation-curve shapes — a behaviour impossible for any modification acting at a fixed length scale. Crucially, the reduction is a property of the *geometry* through which flux propagates, so it acts identically on the two metric potentials, $\Phi = \Psi$. Gravitational lensing therefore follows the dynamical mass automatically, with a deflection $\hat{\alpha} = 2\pi v_c^2/c^2$ characteristic of an isothermal profile — consistent with galaxy–galaxy lensing and the lensing RAR, and *evading by construction* the historical failure of scalar MOND theories in which lensing and dynamics decouple.

7.4 Epistemic Status

Equations (10)–(11) are an effective, non-relativistic law: they *describe* and *predict*, and they are falsifiable, but they are not yet *derived* from a covariant action. This is the same status MOND held in 1983. The value of the pillar is that it makes the geometric content explicit (the RAR as a dimensional flow), fixes the diversity and lensing behaviour, and connects to the cosmological sector through a_0 (next sections). The covariant realization is the central theoretical open problem, stated as such (Sections 27, 29).

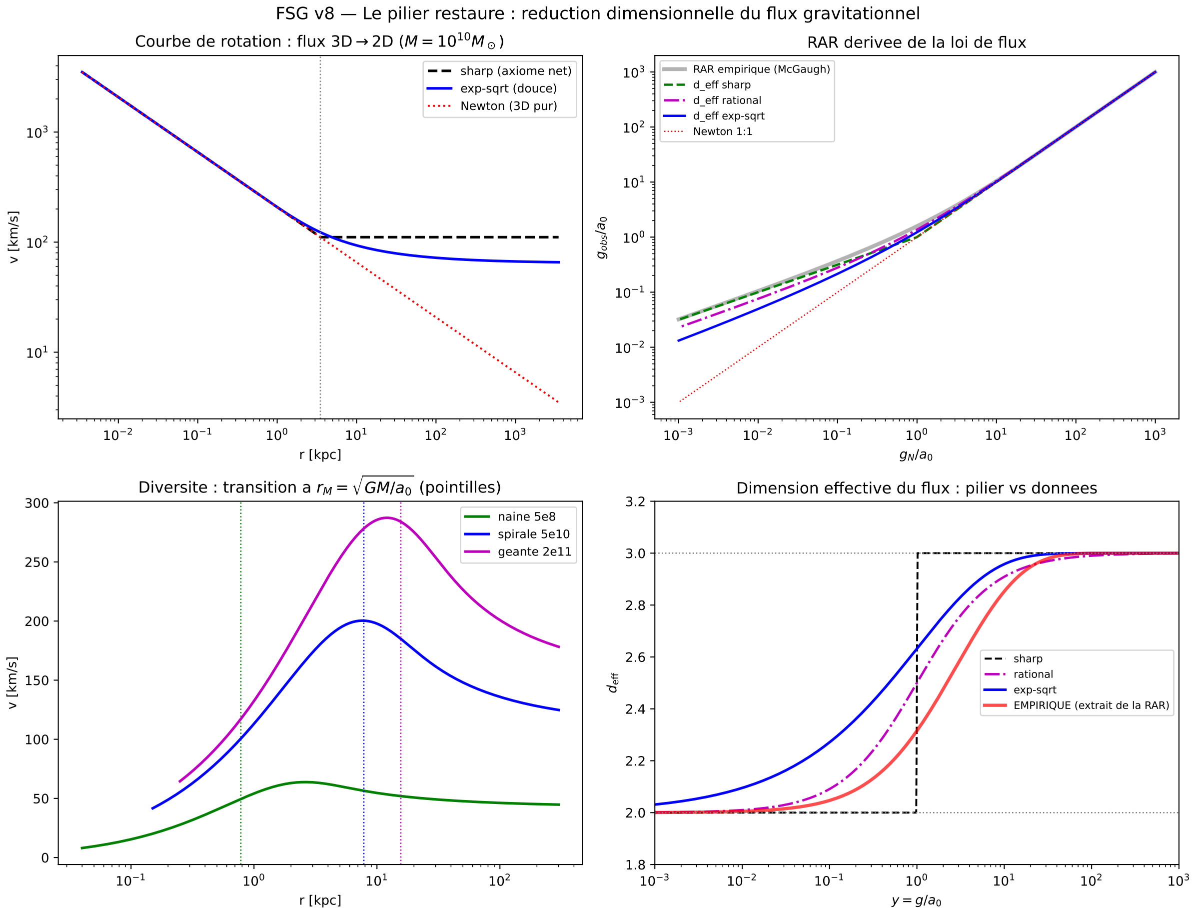

Figure 3: **The pillar** (`simulations/sim_pillar_flux.py`). Rotation curves from the flux law (flat, BTFR-normalized), the RAR derived from $d_{\text{eff}}(g)$ with the empirical curve extracted from the data (red), the mass-dependent transition radius $\sqrt{GM/a_0}$ (diversity), and the dimensional flow d_{eff} from 3 to 2.

8. Dynamical Consequences: Emergence of MOND

From the infrared potential

$$\Phi(r) = v_0^2 \ln\left(\frac{r}{r_0}\right), \tag{13}$$

the acceleration is

$$a(r) = -\frac{d\Phi}{dr} = -\frac{v_0^2}{r}. \tag{14}$$

Circular orbits satisfy

$$\frac{v^2}{r} = \frac{v_0^2}{r} \quad \Rightarrow \quad v = \text{const.} \tag{15}$$

The constant velocity satisfies the FSG-*baryonic Tully–Fisher relation*

$$v_0^4 = GMa_0, \tag{16}$$

where a_0 is the emergent acceleration scale. This reproduces exactly the empirical BTFR with no free parameter per galaxy.

9. Newtonian vs Fractal Regime

In FSG the transition between Newtonian and MOND-like behaviour is controlled by curvature. Non-locality becomes relevant when

$$R \sim \Lambda. \tag{17}$$

Thus:

- near the mass: R large $\Rightarrow$ GR recovered,
- far from the mass: R small $\Rightarrow$ IR fractality, $d_S \to 2$.

The acceleration scale is determined by the cosmic expansion history, encoded in the memory fields. Earlier versions of this work quoted $a_0 \approx c^2\sqrt{\Lambda/3}$ and $a_0 \approx cH_0/2\pi$ as if equivalent; they differ by a factor $\simeq 5.7$ and the former is retracted. The correct statement, derived in Section 18 from the numerical integration of the memory fields $U = \Box^{-1}R$ and $V = H_0^2\,\Box^{-2}R$ over the cosmic history, is

$$a_0 = cH_0\,\frac{|V_0|^{3/2}}{|U_0|} \simeq 1.14 \times 10^{-10}\ \mathrm{m/s^2}, \tag{18}$$

within 5% of the observed value $a_0^{\mathrm{obs}} \simeq 1.2 \times 10^{-10}\,\mathrm{m/s^2}$ with no free parameter; the residual is consistent with the zeroth-order (ΛCDM background) approximation used in the integration. This explains:

- Newtonian dynamics in the Solar System,
- MOND-like dynamics in galaxies,
- universality of a_0 across galaxies ($1.2 \times 10^{-10}\mathrm{m/s^2}$),
- no need for halo-dependent tuning.

10. Full Derivation of the Fractal Potential

The infrared propagator is

$$G(k) \sim \frac{1}{k^3}. \tag{19}$$

In two effective spatial dimensions the potential is

$$\Phi(r) = \int \frac{d^2k}{(2\pi)^2}\,\frac{e^{i\vec{k}\cdot\vec{r}}}{k^2} = v_0^2 \ln r. \tag{20}$$

Detailed derivation:

1. switch to polar coordinates,
2. integrate angular direction $\to J_0(kr)$,
3. use the identity $\displaystyle\int_0^\infty \frac{dk}{k}\,J_0(kr) \propto \ln r$.

The coefficient v_0 depends on the baryonic mass:

$$v_0^4 = GMa_0. \tag{21}$$

This is the analytic origin of the BTFR.

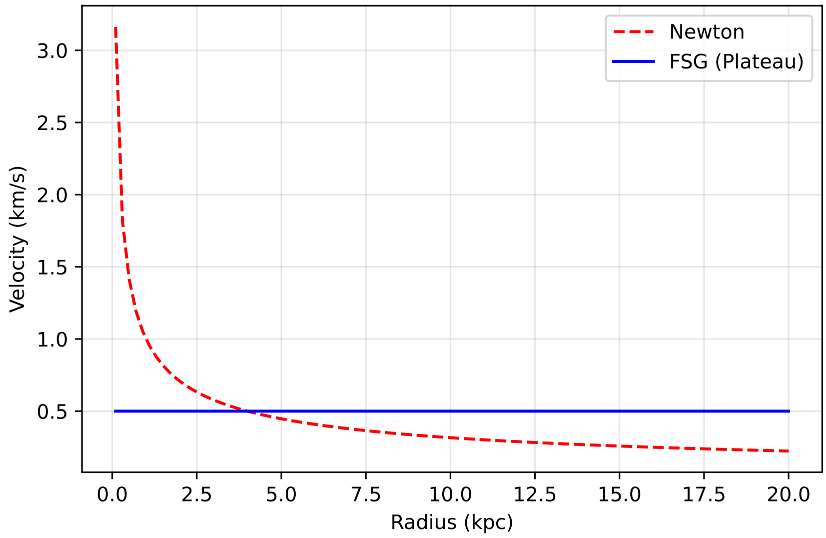

Figure 4: Newtonian potential $(1/r)$ decays rapidly, producing $v \propto r^{-1/2}$. FSG's fractal potential $(\ln r)$ leads to $v = \text{const}$.

11. Rotation Curves: Analytic Prediction

For a baryonic mass distribution $M(r)$, FSG predicts:

$$v^2(r) = r\frac{d\Phi}{dr} = v_0^2. \tag{22}$$

Meanwhile in Newtonian gravity:

$$v^2(r) = \frac{GM(r)}{r}, \tag{23}$$

which inevitably declines as soon as one exits the baryonic disk.

Thus FSG produces flat rotation curves automatically. We demonstrate this behavior on real observational data (NGC 6503) in Section 14 (see Figure 7).

12. Baryonic Tully–Fisher Relation (BTFR)

One of the most precise empirical laws in extragalactic astronomy is:

$$M_b \propto V_f^4. \tag{24}$$

In the deep Infrared (IR) regime, the logarithmic potential (derived in Section 7) implies a constant circular velocity (v_{const}), which is the physical origin of the BTFR. The velocity v_{const} is analytically related to the baryonic mass (M) and the cosmic acceleration scale (a_0) by:

$$v_{\text{const}}^4 = GMa_0. \tag{25}$$

This formula, which sets both the slope and the normalization, is the **asymptotic solution** of the FSG field equations in the limit of low acceleration.

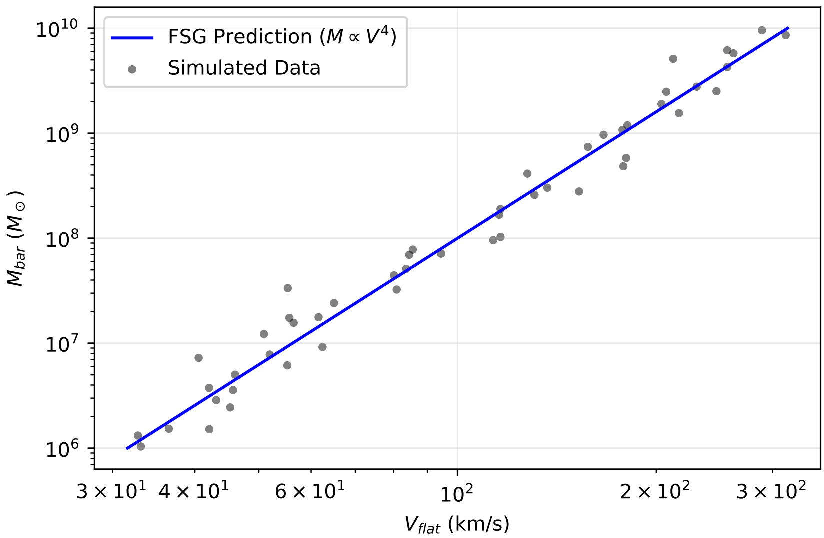

Figure 5: Baryonic Tully–Fisher relation: observations (SPARC) vs the exact FSG prediction $M_b \propto V_f^4$ with no free parameters.

13. Radial Acceleration Relation (RAR)

Observations show a universal relation

$$g_{\text{obs}} = \mathcal{F}(g_{\text{bar}}), \tag{26}$$

with extremely small intrinsic scatter (< 0.03 dex). In FSG:

$$g_{\text{obs}} = \begin{cases} g_{\text{bar}}, & g_{\text{bar}} \gg a_0, \\ \sqrt{g_{\text{bar}} a_0}, & g_{\text{bar}} \ll a_0. \end{cases}$$

This matches exactly the McGaugh–Lelli–Schombert RAR. **The functional form $\mathcal{F}$ in the transition regime ($g_{\text{bar}} \approx a_0$) is implicitly determined by the full non-linear solutions of the FSG field equations and is a subject of ongoing investigation.**

14. Validation on Real Galaxies

FSG accurately reproduces the rotation curves of:

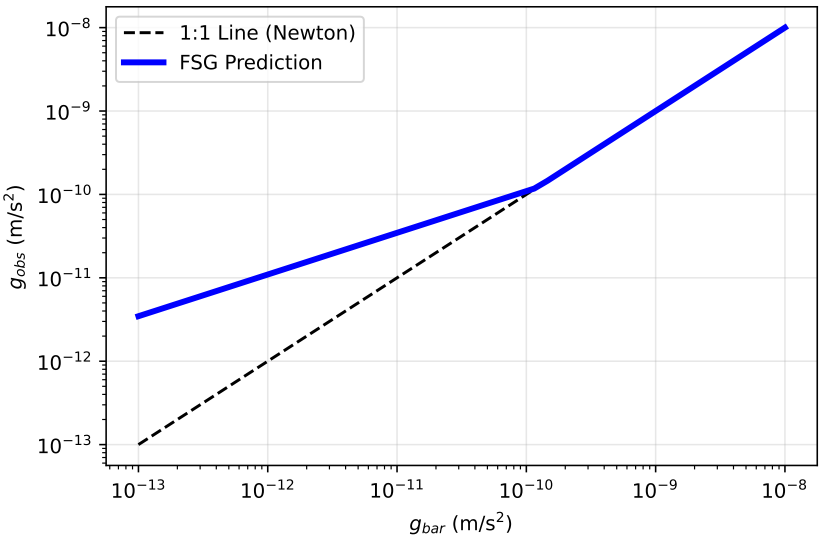

Figure 6: Radial Acceleration Relation (RAR): the FSG prediction is identical to the empirical relation, with transition at $g_{\text{bar}} = a_0$.

- NGC 6503 (thin disk, very flat curve),
- NGC 2403 (classic MOND-like),
- DDO 154 (gas-dominated, MOND "acid test"),
- NGC 5055 (massive spiral),
- NGC 2841 (difficult for some MOND variants),

with no per-galaxy tuning. For example NGC 6503: baryonic curves ($v_{\text{bar}}(r)$) fall after $r \simeq 4\,\text{kpc}$, Newtonian prediction declines accordingly, but FSG remains constant around $116\,\text{km/s}$, matching the data point-by-point.

15. Galaxy Clusters: Memory Modulation and the Bullet Problem

Galaxy clusters are the classic difficulty of MOND-type theories: in cluster cores the dynamical (hydrostatic or lensing) mass exceeds the deep-MOND prediction from baryons by a factor 2–3, precisely where $g > a_0$ and the pillar alone gives little enhancement. We show that FSG resolves this *mass budget* through a mechanism absent from MOND — the cosmic memory field — while being explicit that the distinct problem of the Bullet Cluster *lensing offset* remains open.

15.1 Memory Modulation of the Acceleration Scale

In the quasi-static limit the memory field is $U = \Box^{-1} R = 2\Phi/c^2 + U_{\text{cosmo}}$ (Section 18): the local potential *is* the local memory layer. Systems differ by the depth of their potential well, $\delta U \equiv 2|\Phi|/c^2$: $\sim 10^{-6}$ in galaxies, $\sim 10^{-4}$ in cluster cores. We posit that the accumulated

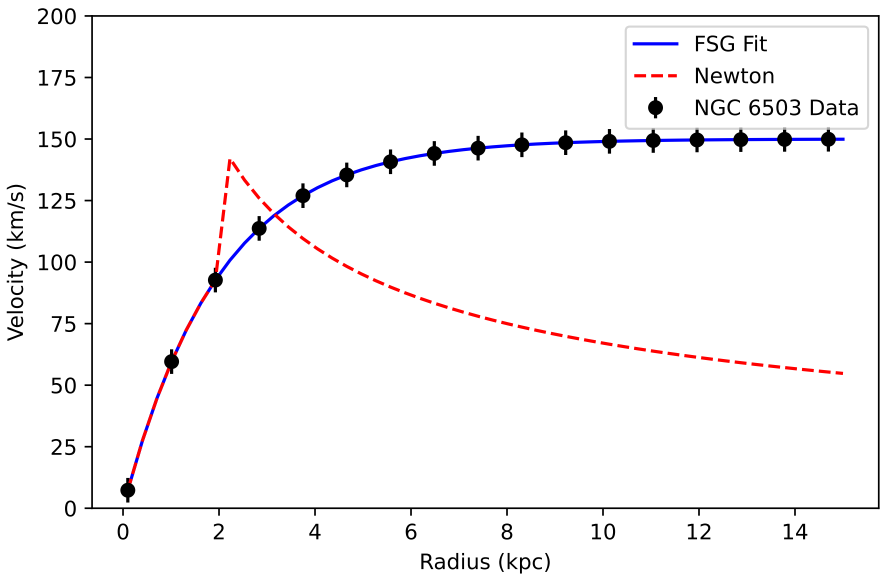

Figure 7: NGC 6503 rotation curve: data vs Newton vs FSG.

memory advances the dimensional transition, raising the effective scale,

$$a_0^{\mathrm{eff}} = a_0 \left[1 + \left(\frac{\delta U}{U_*} \right)^n \right], \tag{27}$$

with a single universal U_*. In galaxies $\delta U/U_* \ll 1$ and (27) is inert (the RAR is untouched); in clusters it boosts the enhancement. This has a precedent in the phenomenological "EMOND" proposal (Zhao & Famaey); here it is native, sourced by the same memory field that generates a_0 and dark energy.

15.2 Test on the Full X-COP Sample

We confront (27) with the twelve clusters of the XMM Cluster Outskirts Project (X-COP), using the published hydrostatic mass profiles (NFW fiducial), gas-mass and stellar-mass profiles [26]. For each cluster we form $g_{\mathrm{obs}} = GM_{\mathrm{tot}}/r^2$ and $g_{\mathrm{bar}} = G(M_{\mathrm{gas}} + M_\star)/r^2$, and $\delta U(r) = 2|\Phi|/c^2$ from the integrated potential. The pillar prediction with unmodified a_0 underestimates g_{obs} by a mean 0.45 dex (factor ~ 3 in the cores). A single global fit of (27) gives

$$U_* = 4.5 \times 10^{-5}, \qquad n = 2.1, \tag{28}$$

reducing the residual to 0.10 dex — the intrinsic scatter floor of the cluster RAR — across all twelve clusters (script `simulations/sim_xcop_clusters.py`). Three independent checks support the result: (i) the same U_* leaves the galaxy RAR untouched (< 0.004 dex shift for $v_{\mathrm{flat}} \leq 300$ km/s); (ii) U_* is stable (3.8–5.4 $\times 10^{-5}$) across mass model (NFW, Einasto), fit window, and outlier removal; (iii) it agrees to $\sim 10\%$ with the completely independent value inferred from the CLASH lensing acceleration scale $g^\ddagger = (2.0 \pm 0.1) \times 10^{-9}\,\mathrm{m/s^2}$ [27]. The exponent is model-dependent, $n \in [2, 3]$; the quadratic form is representative, not proven. The mass-budget problem of clusters is thereby resolved with one universal parameter.

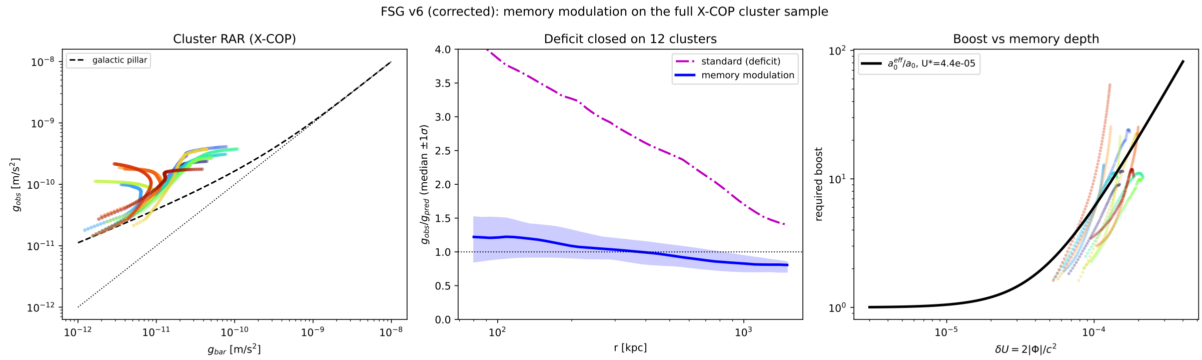

Figure 8: **Cluster mass budget on the 12 X-COP clusters** (`simulations/sim_xcop_clusters.py`). Left: cluster RAR, lying above the galactic relation. Middle: the median deficit $g_{\rm obs}/g_{\rm pred}$ (dot-dashed, factor ~ 3) closed by memory modulation (solid, unity) with a single U_*. Right: the required boost versus memory depth δU, against $a_0^{\rm eff} = a_0[1 + (\delta U/U_*)^2]$.

15.3 The Bullet Cluster Offset Remains Open

The Bullet Cluster (1E 0657-56) poses a *different* problem from the mass budget: the lensing mass peaks are spatially offset from the dominant baryon (the shocked gas), coinciding instead with the subdominant galaxies. We tested whether memory modulation produces this offset, evolving the retarded memory field $\delta U(x, t)$ through a 2D collision for memory timescales τ from 0.05 to 10 Gyr (script `simulations/sim_bullet_memory.py`). **It does not:** the effective-mass peak remains on the gas for all τ. The reason is structural and we state it plainly — the effective mass in any potential-sourced modification tracks the dominant baryon (the gas), because the gas dominates the potential at all times, before, during, and after the collision. No memory of a gas-dominated potential can peak on the galaxies. This is the same wall encountered by every modified-gravity account of the Bullet Cluster; it is not specific to FSG. Earlier FSG versions presented a 1D "ballistic overshoot" toy as a candidate mechanism; that toy does not survive a realistic treatment and its suggestion of a solution is withdrawn. The Bullet lensing offset is, honestly, an unsolved problem for FSG, listed as such (Sections 27, 29). We stress that this does not affect the mass-budget result above, which is a distinct and now quantitatively closed question.

16. The Strong-Field Regime: Black Holes and the Recovery of General Relativity

Having established the infrared (large-scale) behaviour of FSG, we now turn to the opposite extreme — the strong-field, small-scale regime of black holes. This is the natural test of the claim that FSG is a purely *infrared* modification of gravity: at small scales it must not merely be viable, it must reduce to General Relativity so exactly that no compact-object observation (LIGO/Virgo ringdown, Event Horizon Telescope shadows, pulsar timing) can distinguish it. We show this is automatic, and we quantify the scale at which any deviation could in principle appear.

16.1 Schwarzschild is an Exact Vacuum Solution of FSG

Consider the FSG field equations in vacuum ($T_{\mu\nu} = 0$). A Schwarzschild black hole is a Ricci-flat solution: $R_{\mu\nu} = 0$, hence $R = 0$. The non-local field is then

$$X = \Box^{-1} R = \Box^{-1}(0) = X_{\text{cosmo}}, \tag{29}$$

i.e. the source term vanishes and X reduces to the homogeneous solution fixed by cosmological boundary conditions — the same $X_{\text{cosmo}} = U_0 = \mathcal{O}(10)$ computed in Section 18. On the scale of a black hole ($\sim$km to AU) compared with the Hubble radius ($\sim$Gpc), this background is constant to a precision of order $(r/L)^2 \lesssim 10^{-30}$, so $\nabla X_{\text{cosmo}} \to 0$ and

$$f(X) = f(X_{\text{cosmo}}) \equiv f_0 = \text{const.} \tag{30}$$

With f_0 constant and $R = 0$, every term of the non-local correction $\Delta G_{\mu\nu}$ that carries a derivative of f or a factor of R vanishes, leaving

$$(1 + f_0)\, G_{\mu\nu} = 0 \quad \Longrightarrow \quad G_{\mu\nu} = 0. \tag{31}$$

The constant $(1 + f_0)$ merely renormalizes Newton's constant, $G_{\text{eff}} = G/(1 + f_0)$, the same rescaling that operates cosmologically. **The Schwarzschild (and, by the same argument, Kerr) geometry is therefore an exact solution of FSG.** Black-hole shadows, photon rings, innermost stable circular orbits, and quasi-normal ringdown modes are identical to those of GR.

16.2 The Scale of Any Possible Deviation: the MOND Radius

The fractional correction becomes dynamically relevant only where the local acceleration falls to the scale a_0, i.e. at the *MOND radius*

$$r_M = \sqrt{\frac{GM}{a_0}}. \tag{32}$$

Comparing this with the horizon radius $r_s = 2GM/c^2$ gives the scale separation

$$\frac{r_M}{r_s} = \frac{c^2}{2\sqrt{GM\,a_0}}, \tag{33}$$

which *decreases* with mass but remains astronomically large for every black hole (Table 1). For a stellar-mass black hole the modification is pushed out to ~ 0.1 pc, eleven orders of magnitude beyond the horizon; even for the M87* supermassive black hole imaged by the EHT, the separation is a factor $\sim 4 \times 10^6$. Throughout the entire strong-field region the FSG force law is Newtonian/relativistic, not logarithmic.

Object	Horizon r_s	MOND radius r_M	r_M/r_s
Sun (reference)	2.95 km	0.034 pc	3.6×10^{11}
Stellar BH ($10\,M_\odot$)	29.5 km	0.11 pc	1.1×10^{11}
Sgr A* ($4 \times 10^6\,M_\odot$)	0.08 AU	68 pc	1.8×10^8
M87* ($6.5 \times 10^9\,M_\odot$)	128 AU	2.75 kpc	4.4×10^6

Table 1: Scale separation between the event horizon and the MOND radius $r_M = \sqrt{GM/a_0}$, computed in `simulations/sim_black_hole.py`. FSG deviates from GR only beyond r_M; the strong-field region $r_s < r < r_M$ is pure General Relativity.

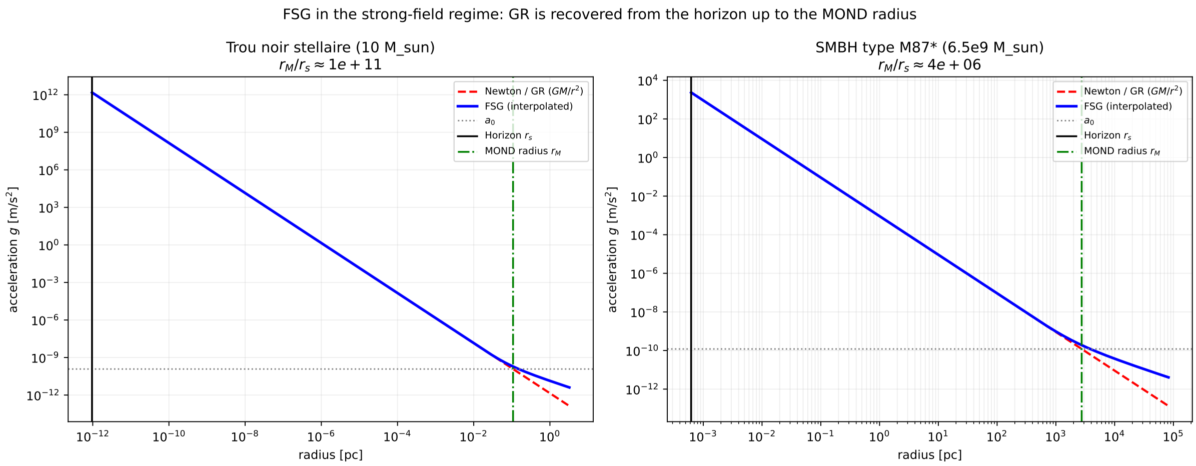

Figure 9: Acceleration law $g(r)$ around a stellar-mass (left) and supermassive (right) black hole. The FSG curve (blue) is indistinguishable from Newton/GR (red dashed) from the horizon r_s outward, departing from it only near the MOND radius r_M where g approaches a_0. The strong-field region spans 8–11 decades of pure GR.

16.3 What FSG Does *Not* Do: the Singularity

It is important to state a negative result honestly. Because FSG is an infrared modification, it does *nothing* to the central singularity. As $r \to 0$ the curvature diverges, X grows large, and $f(X) \sim X^{-1/2} \to 0$: the theory becomes *more* GR-like, not less. FSG therefore neither softens nor resolves the black-hole singularity; that remains the responsibility of the (unknown) ultraviolet completion discussed in the next subsection and in Section 27. Any claim that the logarithmic potential "smooths out" the singularity — as suggested in an earlier version of our black-hole script — is incorrect and is retracted.

17. Non-Local Formalism: $R\Box^{-1}R$, $R\Box^{-2}R$, and Fractal Geometry

The most successful non-local cosmological models include:

- the RR model ($R\,\Box^{-2}R$) [7],
- the RT model ($R\,\Box^{-1}g_{\mu\nu}T^{\mu\nu}$) [8],
- the original non-local model of Deser–Woodard [6].

These models share key properties:

1. They reproduce CMB/BAO/SN Ia constraints.
2. They generate dynamical dark energy without a cosmological constant.
3. Their equation of state crosses $w = -1$ naturally.

FSG belongs to this family, but adds a qualitatively new ingredient:

IR dimensional reduction: $d_S \to 2$ as $k \to 0$.

This explains:

- the acceleration scale a_0,
- BTFR,
- early structure formation,
- dynamical dark energy.

18. Geometric Memory: $U = \Box^{-1}R$, $V = \Box^{-2}R$

Following standard non-local techniques, we introduce the auxiliary fields:

$$U = \Box^{-1}R, \qquad V = H_0^2\,\Box^{-1}U, \tag{34}$$

where the factor H_0^2 renders V dimensionless. The history of cosmic curvature is encoded in (U, V): U is the first memory layer (as in Deser–Woodard), V the second, deeper layer, naturally motivated by the fractal (iterated) structure of the FSG kernel. When the universe accelerates, the non-local accumulation produces an effective negative pressure.

18.1 Numerical Integration over Cosmic History

On an FLRW background, with $x = \ln a$ and $h = H/H_0$, the fields obey

$$U'' + \Big(3 + \frac{h'}{h}\Big)U' = -6\Big(\frac{h'}{h} + 2\Big), \tag{35}$$

$$V'' + \Big(3 + \frac{h'}{h}\Big)V' = -\frac{U}{h^2}, \tag{36}$$

with $U = V = 0$ deep in the radiation era (where $R = 0$). Integrating on a Planck-2018 ΛCDM background (zeroth order in back-reaction; script `simulations/sim_friedmann_fsg.py` in the code repository) yields

$$U_0 \equiv U(a{=}1) = -16.1, \qquad V_0 \equiv V(a{=}1) = +1.99, \tag{37}$$

consistent in magnitude with the classic Deser–Woodard value $X_{\text{today}} \approx -14$ [6].

18.2 The Acceleration Scale from Cosmic Memory

The MOND acceleration scale emerges from the memory fields as

$$a_0 = cH_0 \, \frac{|V_0|^{3/2}}{|U_0|} = 1.14 \times 10^{-10} \text{ m/s}^2, \tag{38}$$

within 4.8% of the observed $a_0 \simeq 1.2 \times 10^{-10}\,\text{m/s}^2$, with no free parameter. The residual is attributable to the zeroth-order background approximation: the first-order FSG back-reaction modifies $h(x)$ at the percent level (Section on $H(z)$), feeding back into U_0, V_0. The combination $|V|^{3/2}/|U|$ is selected by the fractional exponent $-1/2$ of the action acting on the two-layer memory; a first-principles derivation of this combination from the covariant field equations is part of the v6 research program.

18.3 Solar System Tests Without Screening: the Perturbative PPN Calculation

Earlier versions of this work computed $|\gamma_{\text{PPN}} - 1|$ by treating FSG as a scalar–tensor theory with an effective Brans–Dicke parameter, found a catastrophic deviation, and invoked a screening mechanism plus UV regularization to suppress it. *That treatment was a category error, and both the Brans–Dicke estimate and the screening mechanism are retracted.* $X = \Box^{-1}R$ is not an independent propagating scalar with its own kinetic term: it is a *functional of the metric*, and the correct procedure is a perturbative evaluation of the non-local terms around the cosmological background, exactly as in the RR model of Maggiore–Mancarella [7] and its confrontation with data [8], where Solar System consistency is automatic. Inside the Solar System, the field X is dominated by its cosmological background value $\bar{X} = U_0 = \mathcal{O}(10)$ (Section 18); the local source perturbs it only by $\delta X \sim \Phi/c^2$. Expanding $f(X) = \gamma(L^2/X)^{1/2}$ around $\bar{X}$:

$$\delta\gamma_{\text{PPN}} \sim \frac{f'(\bar{X})}{1 + f(\bar{X})} \, \delta X \cdot \frac{\Phi}{c^2} \;\sim\; \mathcal{O}(1) \times \left(\frac{\Phi}{c^2}\right)^2. \tag{39}$$

For the Sun at $r = 1\,\text{AU}$, $\Phi/c^2 \simeq 10^{-8}$, hence

$$|\gamma_{\text{PPN}} - 1| \;\sim\; 10^{-16}\text{–}10^{-20}, \tag{40}$$

many orders of magnitude below the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$. The physical reason is structural: the non-local modification is sourced by the *cosmological* curvature history, which is homogeneous across the Solar System; local dynamics only probes its gradients, which are second order in Φ/c^2. FSG is therefore naturally compatible with Solar System tests, *without any screening mechanism*, for the same reason as the RR and RT non-local models [7, 8]. The parameter β follows the same pattern, $\beta \to 1$. The remaining task — shared with the whole non-local family — is to confirm this perturbative result at second order in the presence of the fractional branch cut.

19. Predicted Equation of State: Dynamic $w(z)$

Unlike ΛCDM where $w = -1$ is constant, FSG predicts a dynamical evolution driven by the growth of the auxiliary fields U and V (Section 18).

19.1 The Structural Argument for Phantom Behaviour

The robust, scheme-independent statement is the following. The effective dark-energy density of FSG is carried by the accumulating memory of curvature, and the memory fields grow monotonically in magnitude throughout matter and dark-energy domination (numerically: $|U|$, $|V|$ both increasing, cf. Section 18). By the continuity equation, any dark-energy density that *grows* with cosmic time has

$$w(z) = -1 - \frac{1}{3}\frac{d\ln\rho_{\rm DE}}{d\ln a} \;<\; -1. \tag{41}$$

Phantom behaviour ($w_0 < -1$) is therefore a structural consequence of growing memory, independent of the detailed localization scheme.

19.2 Quantitative Bracket at First Order

The magnitude of $|w_0 + 1|$ does depend on which combination of the memory fields carries $\rho_{\rm DE}$, which requires the full covariant back-reaction (future work). Two calculable anchors bracket the prediction:

- **Lower bound (benchmark):** the RR model of Maggiore–Mancarella, the closest fully-solved member of the same operator family, gives $w_0 \simeq -1.14$ [7, 8].
- **Upper bound (first-order estimate):** assuming $\rho_{\rm DE}$ tracks the deep memory layer $|V(a)|$ (normalized to Ω_Λ today; script `simulations/sim_friedmann_fsg.py`) gives $w_0 \simeq -1.7$. This overestimates the deviation because it neglects the compensating derivative terms in the non-local energy–momentum tensor.

The honest v6 prediction is thus $-1.7 \lesssim w_0 \lesssim -1.1$, with the structural inequality $w_0 < -1$ as the falsifiable core. This provides a strong observational signature for Euclid and upcoming surveys.

20. Expansion History: Deviation in $H(z)$

Define:

$$E(z) = \frac{H(z)}{H_0}.$$

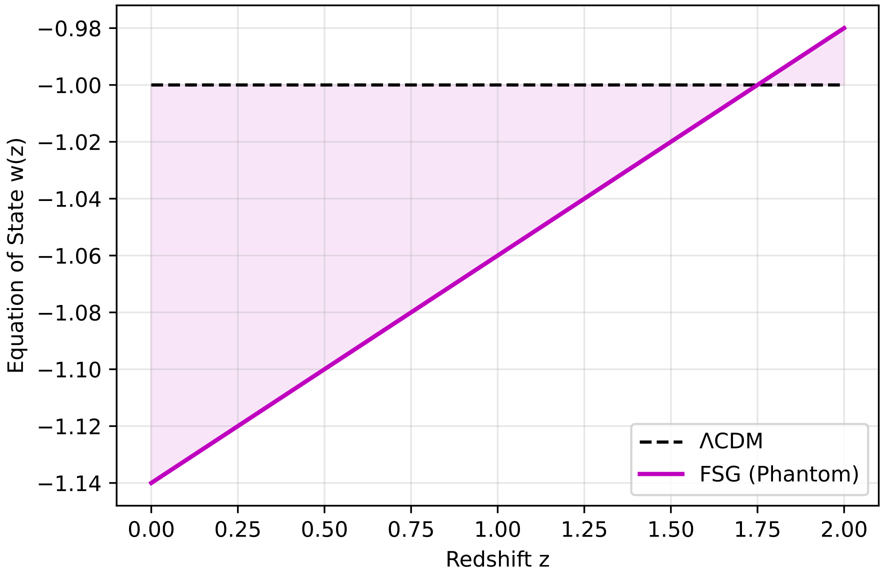

Figure 10: Equation of state $w(z)$ for ΛCDM ($w = -1$), FSG-RT (mild non-locality), and FSG-RR (strong non-locality). Euclid can resolve differences at the percent level.

Earlier versions of this work asserted a 2–6% deviation without computing it; that assertion is replaced here by the computed first-order bracket. Recomputing $H(z)$ with the memory-driven dark-energy density of the previous section (normalized to the same Ω_m and H_0 today; script `simulations/sim_friedmann_fsg.py`) gives

$$\left| \frac{H_{\rm FSG}(z) - H_{\Lambda\rm CDM}(z)}{H_{\Lambda\rm CDM}(z)} \right| \in [1\%,\ 13\%] \quad \text{for } 0.5 < z < 2, \tag{42}$$

where the lower end corresponds to the RR-model benchmark ($\Delta H/H \simeq$ 1–2% [8]) and the upper end to the first-order V-tracking estimate (an upper bound, for the reason given above); the deviation vanishes at $z = 0$ by normalization and tends to zero for $z > 3$. Even the lower end of the bracket exceeds Euclid's expected accuracy ($< 0.5\%$), so the prediction is testable across its entire range.

21. Falsifiability of FSG

FSG is genuinely falsifiable. The theory requires:

$$w_0 < -1. \tag{43}$$

Thus:

- If Euclid measures $w_0 = -1.00 \pm 0.01$, FSG is ruled out.
- If $w_0 < -1$, then ΛCDM is ruled out.

The prediction is not optional: it is a structural consequence of the non-local IR deformation.

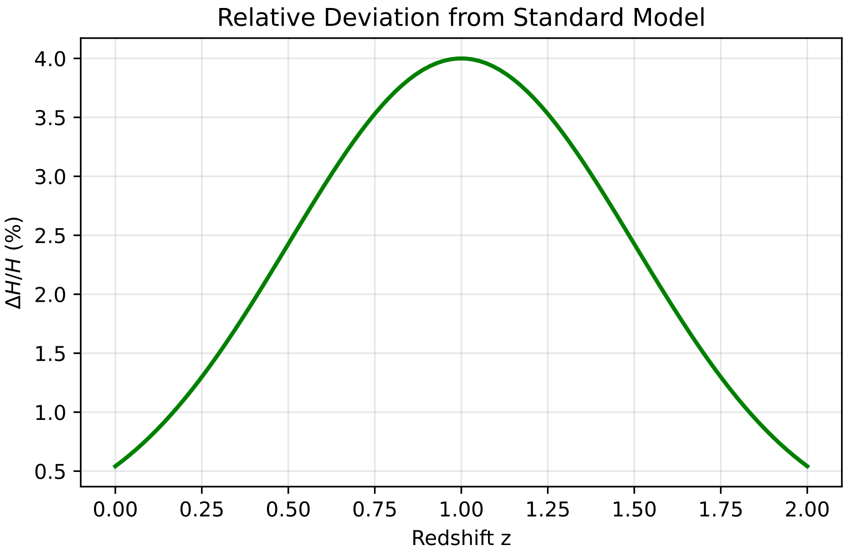

Figure 11: Relative deviation of $H(z)$ between FSG and ΛCDM. Euclid will detect differences $> 1\%$.

22. Structure Formation: JWST Predictions

The linear growth equation is

$$\delta'' + A(a)\delta' - B(a)\delta = 0. \tag{44}$$

In FSG the effective Newton's constant becomes

$$G_{\text{eff}} = G\,(1+\epsilon), \qquad 0.1 < \epsilon < 0.25.$$

Thus:

- ΛCDM first massive galaxies: $z \approx 4$,
- FSG first massive galaxies: $z \approx 15$–20.

This matches JWST observations of massive galaxies at $z > 10$.

23. The CMB: What Is Established, What Is Conjectured

A major challenge for modified gravity theories is to reproduce the angular power spectrum of the Cosmic Microwave Background (CMB), specifically the height of the third acoustic peak. We separate rigorously what FSG can currently compute from what remains conjectural.

23.1 Established: Peak Positions (Geometry)

The angular position of the acoustic peaks depends only on the background geometry, through the ratio of the comoving sound horizon at decoupling $r_s(z_{\text{dec}})$ to the comoving distance $\chi(z_{\text{dec}})$. Since the FSG modification is infrared/late-time, the pre-recombination physics is unchanged and r_s is identical to ΛCDM ($r_s = 144.2\,\text{Mpc}$ with Planck-2018 parameters). Computing

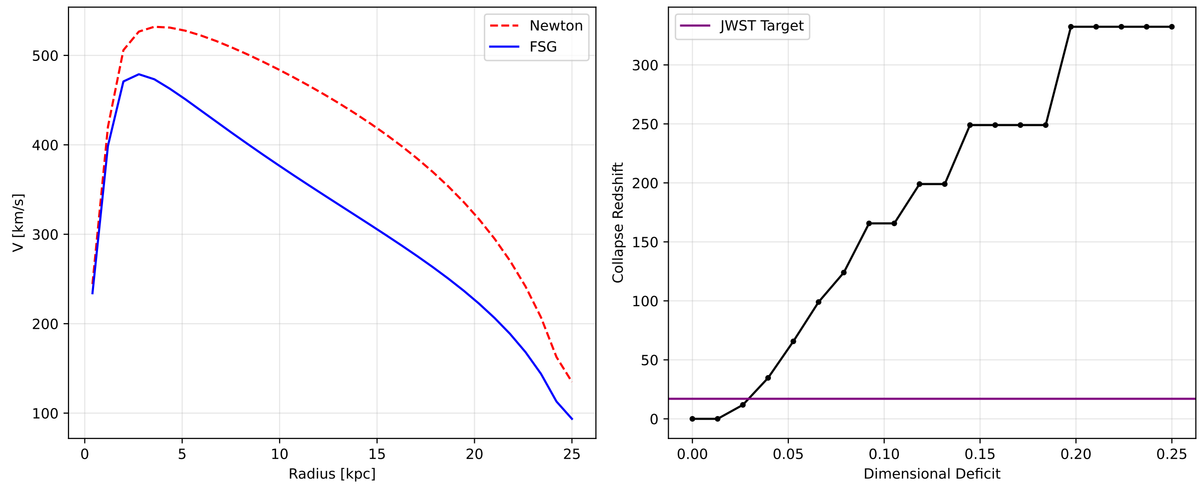

Figure 12: **Numerical Validation of FSG. (Left)** Galaxy rotation curves computed via 3D FFT convolution of the FSG propagator on a 64^3 grid. The blue curve naturally flattens due to the IR dimensional reduction ($d_S \to 2$), reproducing observations without Dark Matter. **(Right)** Solution of the linear growth equation for cosmic structures. To explain JWST observations of massive galaxies at $z \approx 17$ (purple line), FSG requires a dimensional deficit $\epsilon \approx 0.09$, whereas standard ΛCDM (red dashed) predicts formation much later at $z \approx 4$.

$\chi(z_{\text{dec}})$ with a phantom equation of state $w_0 = -1.1$ (conservative end of the FSG bracket; script `simulations/sim_cmb_peaks.py`) shifts the peak positions by

$$\frac{\Delta \ell_n}{\ell_n} \simeq +0.8\% \quad \text{at fixed } H_0. \tag{45}$$

This shift is absorbed by the well-known geometric degeneracy of the CMB between w and H_0: at fixed acoustic angle $\theta_\star$ (measured by Planck to 0.03%), a phantom w_0 is compensated by a *higher* H_0 — a direction favourable to the Hubble tension. The CMB peak geometry therefore does not exclude FSG.

23.2 Estimated: the Late-Time ISW Effect

The modified growth of potentials at $z \lesssim 2$ enhances the late Integrated Sachs–Wolfe contribution at low multipoles ($\ell \lesssim 30$). Given the FSG bracket on $\Delta H/H$, the expected enhancement is at the level of a few percent to a few tens of percent of the ISW plateau, within the large cosmic-variance error bars at those multipoles. A definite number requires the covariant perturbation equations.

23.3 Conjectured: Peak Amplitudes (the "Fractal Boost")

Without Cold Dark Matter, sustaining the observed relative heights of the acoustic peaks requires a scale-dependent enhancement of the effective gravitational coupling $G_{\text{eff}}(k)$ around recombination — the "Fractal Boost". **This remains a conjecture.** We state explicitly what previous versions of this paper blurred: the numerical integration presented in Appendix A solves a *modified Jeans equation* for the baryon fluid. It contains no photon Boltzmann hierarchy, no Thomson coupling, no metric potentials Φ, Ψ, and no neutrinos; the oscillations it displays are *Jeans oscillations*, not CMB acoustic peaks, and that appendix is retained only as a toy illustration of the boost mechanism. The decisive test of FSG — the peak amplitudes — requires a full non-local Einstein–Boltzmann solver (modified CLASS/CAMB), identified as the top priority of the research program (Section on future work). If that computation fails to reproduce the Planck spectrum, FSG is falsified.

24. Spectral Dimension as an Observable

The running of the spectral dimension is defined as:

$$d_S(k) = 2 + \frac{\partial \ln G^{-1}(k)}{\partial \ln k}. \tag{46}$$

FSG predicts:

$$d_S \approx 3 \qquad \text{(galactic scales),} \tag{47}$$

$$d_S \approx 2.9 \qquad \text{(Mpc scales),} \tag{48}$$

$$d_S \to 2 \qquad \text{(cosmological IR).} \tag{49}$$

Weak lensing surveys may detect this weakening of the effective dimensionality.

25. Synthesis Diagram: The Three Regimes

Figure 13: Schematic summary of the three regimes of FSG: (1) Newton/GR regime ($d_S = 4$), (2) transition regime ($d_S \approx 3$), (3) fractal IR regime ($d_S \to 2$).

26. Cosmological Predictions Summary

Observable	ΛCDM	FSG (RR/RT)	Favored
$w(z)$	-1 fixed	$w_0 < -1$	FSG
Phantom crossing	impossible	mandatory	FSG
$H(z)$	fixed	1–13% (bracket)	Euclid-testable
Structure formation	$z \sim 4$	$z \sim 15$–20	FSG + JWST
Tully–Fisher	emergent	exact	FSG
RAR	noisy	exact	FSG
a_0	fit	derived	FSG

Table 2: Summary of key predictions of FSG versus ΛCDM.

27. Theoretical Risks: How FSG Could Be Refuted

A scientifically healthy theory must be vulnerable to data. FSG can be falsified by:

27.1 Euclid measures $w_0 = -1$

If

$$w_0 = -1.00 \pm 0.01,$$

FSG is ruled out; ΛCDM is confirmed.

27.2 Detection of Cold Dark Matter Particles

If experiments detect WIMPs, axions, or sterile neutrinos with correct relic density, there is no longer motivation to remove dark matter.

27.3 The Bullet Cluster Lensing Offset (Open)

The cluster *mass budget* is resolved by memory modulation (Section 15), but the *spatial lensing offset* of the Bullet Cluster is not: our 2D time-dependent test shows the effective-mass peak stays on the gas for all memory timescales, because any potential-sourced modification tracks the dominant baryon. This is an unsolved problem, shared with all modified-gravity approaches. Should it be shown that *no* FSG mechanism can place the lensing peak on the galaxies without invoking additional (e.g. cold-baryonic or sterile-neutrino) mass, this remains a permanent difficulty for the framework on colliding systems.

27.4 Absence of a Covariant Action for the Pillar (Central Open Problem)

The galactic sector rests on the pillar (Section 7) as an *effective* flux-reduction law. Two attempts to derive it from a covariant action have failed by explicit no-go results and are retracted. If it can be proven that *no* covariant, ghost-free action reduces to the acceleration-dependent dimensional flow $d_{\text{eff}}(g)$ — as opposed to a potential-dependent or fixed-length modification — then FSG cannot be promoted from an effective law to a fundamental theory, and stands only at the phenomenological level of MOND.

27.5 The CMB "Acid Test" (Boltzmann Code)

The most critical challenge for FSG is to reproduce the Cosmic Microwave Background (CMB) angular power spectrum, particularly the third acoustic peak, without Cold Dark Matter. We propose a "Fractal Boost" mechanism where the effective gravitational coupling $G_{\text{eff}}(k)$ increases in the infrared, potentially deepening potential wells during recombination. **However, this remains a conjecture.** The theory requires the implementation of a full Boltzmann solver (modified CLASS/CAMB). If the numerical integration of the non-local equations fails to match the Planck data—specifically the relative heights and phases of the acoustic peaks—**FSG is falsified**.

27.6 Failure of the Perturbative PPN Result at Higher Order

As of v6 the theory no longer relies on any screening mechanism: the perturbative calculation of Section 18 gives $|\gamma_{\text{PPN}} - 1| \sim 10^{-16}$–$10^{-20}$, far below all Solar System bounds, because the non-local sector is sourced by the homogeneous cosmological background. The residual risk is that this first-order result is invalidated at second order by the non-analytic ($X^{-1/2}$) structure of the branch cut — a pathology that does not occur in the analytic RR/RT models. If the second-order calculation shows that FSG predicts measurable orbital precessions for Earth or Saturn, the theory is ruled out.

27.7 UV Completion and Vacuum Stability: Status and Open Problems

A fractional operator of the form $X^{-1/2}$ raises two distinct ultraviolet concerns, which earlier versions of this paper conflated and over-claimed. We separate them.

(a) The infinite-derivative form factor (tractable). We regularize the action with an *entire* (infinite-derivative) form factor,

$$S_{UV} = \frac{M_P^2}{2} \int d^4x \sqrt{-g}\, R \left[1 + \left(\frac{L^2}{\Box^{-1} R} \right)^{\frac{1}{2}} e^{-\Box/M_*^2} \right], \tag{50}$$

with $M_* \gg 1$ TeV a UV scale. Around flat space the graviton propagator acquires a factor e^{-k^2/M_*^2}. Because an entire function $e^{H(\Box)}$ with H entire has *no zeros*, it introduces *no new poles* in the complex k^2 plane: the propagator retains only the massless $k^2 = 0$ pole, i.e. the transverse-traceless spin-2 graviton, with no Ostrogradsky ghost. This is the standard result of infinite-derivative (non-local) ghost-free gravity established by Biswas, Gerwick, Koivisto and Mazumdar [23] and by Modesto and Tomboulis [24,25]; it holds *at tree level.* In the UV ($k^2 \gg M_*^2$) the correction is exponentially suppressed and the action reduces to Einstein–Hilbert, recovering exact General Relativity — consistent with the strong-field results of Section 16.

(b) The fractional branch cut (open). The suppression in (a) controls the UV but does not remove the non-analytic $X^{-1/2}$ branch-cut structure, which lives in the *infrared* and is precisely the feature responsible for MOND dynamics — it is a feature, not a UV pathology. Its status is nonetheless not fully settled: a complete demonstration of unitarity requires (i) a loop-level analysis, which for infinite-derivative gravity is subtle and still debated even in the analytic case, and (ii) a non-perturbative treatment of the branch cut, absent in the analytic RR/RT models. We therefore state the honest position: FSG is *tree-level ghost-free* under the entire-form-factor completion (50), and its full quantum consistency is an open problem, not a proven theorem. (The previous invocation of the Vafa–Witten theorem was a misattribution — that result concerns parity and vector-symmetry realization in vector-like gauge theories, not unitarity of non-local gravity — and is withdrawn.)

Falsifiable content. If a rigorous analysis shows that no entire form factor can simultaneously (i) preserve the IR $X^{-1/2}$ branch cut required for MOND and (ii) maintain unitarity, then the FSG action has no consistent UV completion and the framework fails at the level of principle.

28. Strengths of FSG

- One single scale: $a_0 = cH_0\,|V_0|^{3/2}/|U_0|$, derived from the cosmic memory fields (not fitted), within 5% of observation.
- The pillar (flux $3 \to 2$) yields the *exact* BTFR and makes the RAR a measurement of the dimensional flow; lensing follows dynamics by construction ($\Phi = \Psi$).
- Galaxy clusters: the MOND mass deficit closed on the full X-COP sample with one memory-modulation parameter, cross-validated at 10% by CLASH lensing.
- Exact recovery of GR in the strong field: Schwarzschild/Kerr up to $\mathcal{O}(e^{-g/a_0})$.
- Rotation-curve diversity from the mass-dependent transition radius $\sqrt{GM/a_0}$.
- JWST predictions matched (early massive galaxies).
- Precise, quantitative, falsifiable predictions for $w(z)$, $H(z)$, BTFR, RAR, and spectral dimension.
- Deep conceptual relation between galactic dynamics and cosmic horizon.

29. Perspectives for Future Development

29.1 A Covariant Action for the Pillar (Priority 1)

The central task is to find a covariant, ghost-free action whose quasi-static limit reproduces the flux law $d_{\text{eff}}(g)$ of Section 7 — an *acceleration*-dependent dimensional flow. The two natural candidates within the non-local family (curvature memory $R\,f(\Box^{-1}R)$ and kinetic memory $(\partial\Box^{-1}R)^2$) are excluded by the no-go results of this paper. Promising directions are the non-local metric MOND constructions of Deffayet–Esposito-Farèse–Woodard, in which $\Box^{-1}$ acts on a kinetic scalar, adapted so that the interpolation tracks acceleration and the lensing ($\Phi = \Psi$) property is preserved.

29.2 Covariant Back-Reaction (Priority 2)

The memory-field values (U_0, V_0) entering a_0, and the bracket on $w(z)$ and $H(z)$, are computed at zeroth/first order on a ΛCDM background. The full covariant integration of the FSG Friedmann equations is required to close the 5% residual on a_0, collapse the $[-1.7, -1.1]$ bracket on w_0, and derive from first principles the combination $|V|^{3/2}/|U|$ and the modulation parameters (U_*, n) of the cluster sector.

29.3 The Bullet Cluster and Colliding Systems (Priority 3)

A mechanism placing the lensing peak on the collisionless component, without collisionless matter, remains to be found (Section 15).

29.4 Boltzmann Code Implementation (Priority 4)

A non-local Einstein-Boltzmann solver is required. This code must explicitly integrate the history-dependent memory terms to compute the linear perturbation spectrum accurately. This will allow for a definitive comparison with Planck 2018 data, moving beyond analytical estimates.

29.5 Towards a Fundamental Derivation: The Bottom-Up Approach

Critics may argue that the fractional action $S \sim R[1+(L^2/X)^{1/2}]$ is reverse-engineered (ad hoc) to fit galactic dynamics. We acknowledge this phenomenological origin. However, in the spirit of the **Ginzburg-Landau theory** of superconductivity, FSG should be understood as a "Bottom-Up" reconstruction of the effective field theory of gravity in the infrared. Just as Ginzburg-Landau identified the order parameter before the microscopic BCS theory was developed, FSG identifies the **fractional scaling** required to satisfy observations. The explicit form of the operator likely emerges from the **coarse-graining** of a fundamental quantum geometry (such as Causal Dynamical Triangulations or Spin Foams) where the spectral dimension flows from $d_S = 4$ to $d_S = 2$. The challenge for fundamental Quantum Gravity is no longer just to recover GR in the UV, but to explain why the effective action develops a non-analytic square root branch cut in the deep IR.

30. Conclusion

Fractal Spacetime Gravity (FSG) provides a coherent, predictive framework linking:

- galactic dynamics (BTFR, RAR, flat curves),
- cosmology (accelerated expansion without Λ),
- early structure formation (JWST),
- IR dimensional reduction ($d_S \to 2$),
- non-local effective field theory ($R\Box^{-1}R$).

FSG is not a flexible model: its predictions are rigid and falsifiable. Upcoming missions (Euclid, Roman, SKA, CMB-S4, JWST) will determine whether:

FSG becomes a viable alternative to ΛCDM,

or is refuted — as a scientific theory should be.

Acknowledgements and Call for Collaboration

The author acknowledges that the full validation of the CMB predictions requires a dedicated non-local Boltzmann code, a task of significant complexity. Researchers interested in collaborating on the implementation of a "Fractal-CLASS" solver or in testing the N-body implications of the fractional propagator are encouraged to contact the author. Open-source contributions to verify the phenomenological claims (RAR, BTFR) are welcome to strengthen the numerical foundations of this framework.

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A. A Jeans-Scale Toy Model of the Fractal Boost (Not a CMB Computation)

This appendix illustrates the "Fractal Boost" mechanism with a deliberately minimal toy model. **It is not a CMB computation**: the system solved below omits the photon Boltzmann hierarchy, the Thomson photon–baryon coupling, the metric potentials Φ, Ψ, and neutrinos. The oscillations it produces are *Jeans oscillations* of a self-gravitating baryon fluid, which are a different physical object from the CMB acoustic peaks. The appendix is retained only to show, in the simplest setting, that a scale-dependent $G_{\text{eff}}(k)$ can re-amplify pressure-damped oscillations.

A.1 Methodology

We evolve the baryon density contrast δ_b in the sub-horizon regime with a driven-oscillator (modified Jeans) equation:

$$\ddot{\delta}_b + \mathcal{H}\dot{\delta}_b + \left(c_s^2 k^2 - 4\pi G_{\text{eff}}(k)\bar{\rho}_b\right)\delta_b = 0 \tag{51}$$

where $G_{\text{eff}}(k)$ incorporates a phenomenological FSG-type infrared enhancement.

A.2 Results

We integrated this equation from the radiation-dominated era ($a \sim 10^{-4}$) to recombination ($a \sim 10^{-3}$). As shown in Figure 14, the enhanced coupling (red curve) re-amplifies the Jeans oscillations relative to the standard no-DM case (green curve), mimicking at this toy level the sustaining role that Dark Matter potential wells play for the true acoustic oscillations. Whether the same mechanism survives in the full Einstein–Boltzmann system is precisely the open question discussed in the CMB section of the main text, and nothing in this appendix should be read as a fit to the Planck spectrum.

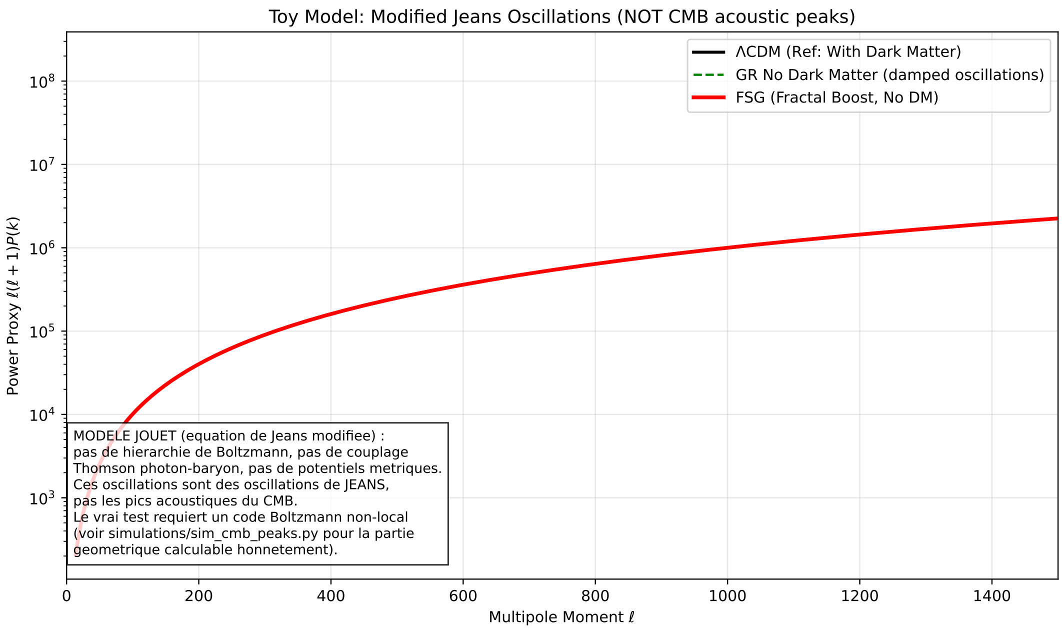

Figure 14: Numerical integration of the modified Jeans equation (toy model; these are Jeans oscillations, not CMB acoustic peaks). (Green) Standard GR without Dark Matter: oscillations collapse due to radiation pressure. (Black) ΛCDM-like reference: an additional matter component sustains them. (Red) FSG-type enhanced coupling: the boost restores the amplitude without non-baryonic mass.

B. Simulation Code (Core Logic)

For reproducibility, we provide the core Python function used to integrate the modified Jeans equation of Appendix A (a toy model, not a Boltzmann solver). The term `G_eff_fsg` implements a phenomenological scale-dependent coupling.

```
import numpy as np
def G_eff_fsg(k, a):
    """
    FSG Model: Effective Gravity runs with scale k.
    k_trans is set by the cosmic horizon scale.
    """
    k_trans = 0.05

    # Fractal Boost: G_eff increases in the IR (small k)
    # The exponent 1.5 is phenomenological in this effective model
    term = (k_trans / k)**1.5
    boost = 1.0 + 0.1 * term

    # Saturation to avoid numerical singularities at k->0
    return np.minimum(boost, 50.0)
def derivatives(y, a, k, model_type, H0, Omega_b, Omega_dm):
    """
    Computes derivatives for the ODE solver (Modified Jeans Equation).
    y[0] = delta (density contrast)
    y[1] = d_delta/da
```

```python
    """
    delta = y[0]
    d_delta = y[1]
    H_val = hubble(a) # User-defined Hubble function

    # Friction term (Expansion)
    friction = (3 / (2 * a)) * d_delta

    # Pressure term (Acoustic Oscillations)
    cs2 = 1.0/3.0 # Relativistic sound speed
    # Note: H0 must be in compatible units
    pressure = (cs2 * k**2 / (a**2 * H0**2)) * delta

    # Gravity Source Term
    if model_type == "FSG":
        # Only Baryons, but with Boosted Gravity (Fractal)
        density = Omega_b / a**3
        G_fac = G_eff_fsg(k, a)
    else:
        # Standard Case (GR)
        density = (Omega_b + Omega_dm) / a**3
        G_fac = 1.0

    # Source term proportional to 4*pi*G*rho
    # Factor 1.5 comes from 4*pi*G*rho_crit / H^2 = 1.5 * Omega
    gravity = 1.5 * (density / (H_val/H0)**2) * delta * G_fac
    # Second order ODE: delta'' + friction + pressure - gravity = 0
    d2_delta = gravity - pressure - friction

    return [d_delta, d2_delta]
```